

### 3 Strategy

#### 3.1 Convergence

For each visibility matrix, the definition for convergence will be the same as for the original riddle. That is, when every blue-eyed agent knows their own eye colour and has left, the riddle has converged. This means that, in order for an agent  $i$  to leave, all the worlds that it considers possible must be worlds in which  $b_i$  is true. Our generalization allows us to examine if convergence still occurs within a finite number of days under partial visibility. Moreover, we can check whether convergence is possibly postponed by partial visibility.

#### 3.2 Original Riddle Example

We illustrate how the classic version of the Blue-Eyed Islanders riddle unfolds. For the previously described Kripke model  $M$ , let us take the example where there are three islanders, so  $a_0$ ,  $a_1$  and  $a_2$ . The set of possible states  $S$  will then be given by  $S = \{0, 1\}^3$  and can be visualized as in Figure 1.

In this visualization, reflexive and symmetric relations are not drawn but they are of course present. Assume that agent  $a_1$  and agent  $a_2$  have blue eyes, meaning that world  $w = 011$  is the true world (underlined in Figure 1). Now, the first public announcement is given that “ $b_0 \vee b_1 \vee b_2$ ”. This restricts the model by removing the world that no islander has blue eyes ( $w = 000$ ). In the remaining worlds, all agents are still uncertain about what the actual configuration should be. Thus, at first dawn, nobody left. This inaction is noted by the agents and can be modeled as the public announcement “ $\neg K_0 b_0 \wedge \neg K_1 b_1 \wedge \neg K_2 b_2$ ”. The fact that nobody is certain, reveals that all agents must observe at least one other agent with blue eyes, while still contemplating whether they themselves have blue eyes. Consequently, the model is once again restricted by removing all states with only one blue-eyed islander (Figure 2). After this inaction,  $w = 011$  is the only world for  $a_1$  and  $a_2$  in line with their observations, while  $a_0$  is still uncertain about the situation. Thus,  $a_1$  and  $a_2$  leave at the second dawn and “ $K_1 b_1 \wedge K_2 b_2$ ” can be concluded.

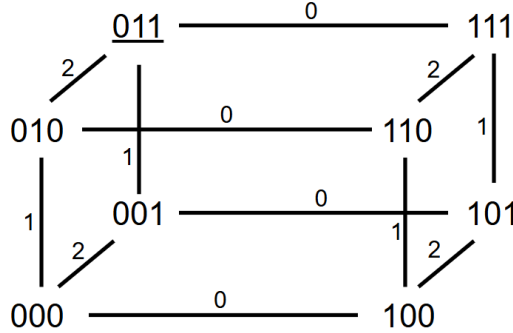


Figure 1: Formal Kripke model for an original Blue-Eyed Islander riddle with 3 agents. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present.

#### 3.3 Converging Example

Our implementation will require a system that, given an amount of agents and fixed blue-eyed islanders, iterates over all possible visibility matrices and evaluates whether such a scenario can

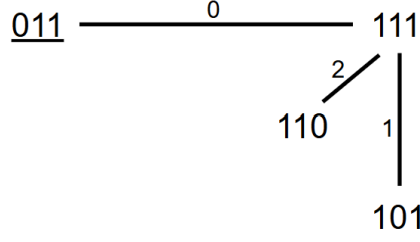


Figure 2: Formal Kripke model for an original Blue-Eyed Islander riddle with 3 agents after the observed inaction at the first dawn. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present.

converge. For clarity, this example will only go over one visibility matrix (similar to the previous example), just to show how the imperfect visibility can alter the accessibility relations and thus the Kripke models. Moreover, it shows the concept of what will happen iteratively during a run of our implementation.

Once again, take 3 agents  $a_0$ ,  $a_1$  and  $a_2$ .  $a_1$  and  $a_2$  have blue eyes, making  $w = 011$  the actual world. However, different from the previous example, visibility is not complete and the visibility matrix  $E = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ . So,  $a_1$  cannot see  $a_0$ , while the rest of the visibility remains the same. The Kripke model for this configuration is illustrated in Figure 3.

It is important to realize that it is common knowledge that another agent cannot see something, as the agents can use this information to their advantage. More specifically, the visibility matrix is assumed as common knowledge throughout our version of the riddle. Now on day 1, the public announcement “ $b_0 \vee b_1 \vee b_2$ ” is given. This removes world  $w = 000$ . Based on this announcement and what everyone can see,  $a_0$  still considers the set of worlds  $\{011, 111\}$  to be possible,  $a_1$  considers  $\{001, 011, 101, 111\}$  to be possible, and  $a_2$  considers  $\{010, 011\}$  to be possible.

The next dawn, nobody left. Now, everyone can remove the worlds which would see one of the agents leave. This includes world  $w = 100$  ( $a_0$  could have left since it has perfect vision), but nobody did even consider this one. Consequently,  $a_1$  does no longer consider  $w = 001$  to be possible, since  $a_2$  would have left if he was the only blue-eyed islander. Thus,  $a_2$  must see that at least one other islander has blue eyes, so  $a_1$  is left with the set  $\{011, 101, 111\}$ .  $a_2$  cannot do the same for  $w = 010$ , because he knows that  $a_1$  has imperfect vision and thus concludes that even if only  $a_1$  has blue eyes,  $a_1$  would not be able to leave. The situation after the first dawn can be described by the Kripke model in Figure 4.

The next dawn, nobody left.  $a_1$  takes this information and is able to deduce that itself must have blue eyes and is left with the set  $\{011, 111\}$ .  $a_1$  can remove  $w = 101$  because it realizes that  $a_0$  and  $a_2$  would have left if  $w = 101$  was the actual world. Assuming that  $w = 101$  is the actual world,  $a_0$  and  $a_2$  would only be uncertain about their own eye colour. This makes it that they could not leave at the first dawn. However, since nobody left, they would have the information necessary to leave at the second dawn. Since this did not happen,  $a_1$  could remove this possible world and conclude that itself must have blue eyes.

Moreover,  $a_2$  can exclude  $w = 010$  for similar reasoning. Assuming  $w = 010$  is the actual world, we know that  $a_1$  could not leave at the first dawn since it has partial vision and cannot

see  $a_0$ 's eye colour (still considers  $\{010, 110, 100\}$ ). We also know that  $a_0$  could not leave because it cannot see its own eye colour (still considers  $\{010, 110\}$ ). From this information,  $a_1$  would exclude  $w = 100$  because  $a_0$  would have left otherwise. So,  $a_1$  only considers the set  $\{010, 110\}$  and thus should leave the next dawn. Nonetheless, it did not leave, providing  $a_2$  the information that  $w = 010$  cannot be the actual world and thus that the actual world must be  $w = 011$ . The Kripke model after these restriction is illustrated in Figure 5.

Therefore,  $a_1$  and  $a_2$  were able to leave at the third dawn, meaning that it took one dawn more to converge compared to the classical riddle for 2 blue-eyed islanders.

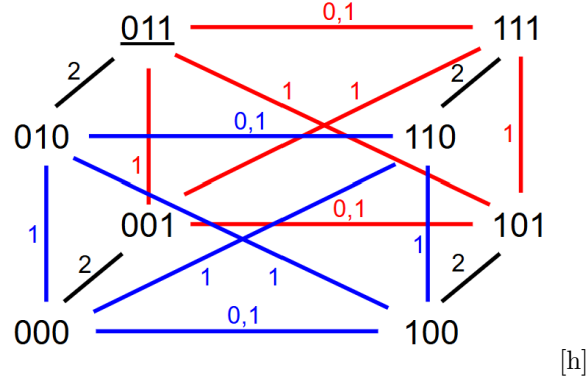


Figure 3: Example Kripke model for the Blue-Eyed Islander riddle with a partial visibility configuration (agent 1 cannot see agent 0). The actual world is underlined. Reflexive and symmetric relations are not drawn but are present. Colours in the model only serve visual purposes, as the partial vision models can become difficult to interpret.

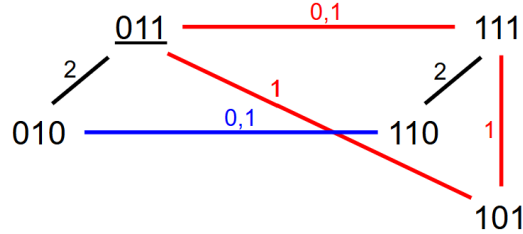


Figure 4: Formal Kripke model for the Blue-Eyed Islander riddle with a partial visibility configuration (agent 1 cannot see agent 0) after the first dawn. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present. Colours in the model only serve visual purposes, as the partial vision models can become difficult to interpret.

### 3.4 Non-converging Example

Take agents  $a_0$ ,  $a_1$  and  $a_2$ . Once again,  $w = 011$  is the actual world. In this example, the visibility matrix  $E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ . So,  $a_0$  and  $a_1$  cannot see each other. The Kripke model for this configuration is illustrated in Figure 6.

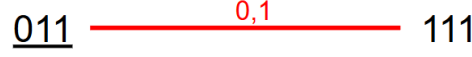


Figure 5: Formal Kripke model for the Blue-Eyed Islander riddle with a partial visibility configuration (agent 1 cannot see agent 0) after the second dawn. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present. Colours in the model only serve visual purposes, as the partial vision models can become difficult to interpret.

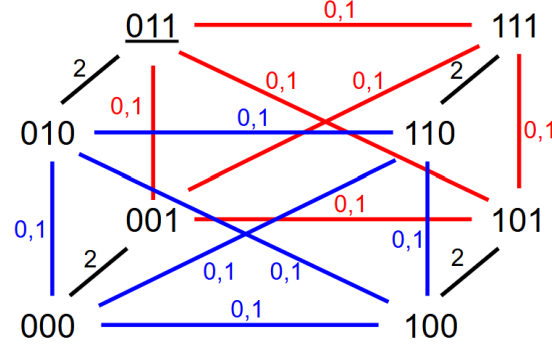


Figure 6: Example Kripke model for the Blue-Eyed Islander riddle with a partial visibility configuration (agent 0 and agent 1 cannot see each other). The actual world is underlined. Reflexive and symmetric relations are not drawn but are present. Colours in the model only serve visual purposes, as the partial vision models can become difficult to interpret.

After the announcement “ $b_0 \vee b_1 \vee b_2$ ”, world  $w = 000$  can be excluded. Based on this announcement and what everyone can see, both  $a_0$  and  $a_1$  consider the set of worlds  $\{001, 011, 101, 111\}$  to be possible, and  $a_2$  considers  $\{010, 011, 100\}$  to be possible.

The next dawn, nobody left. Assuming  $w = 001$ ,  $a_2$  would have left. Therefore,  $a_0$  and  $a_1$  no longer consider it.  $a_2$  still has to consider  $w = 100$  and  $w = 010$ , because it knows that both  $a_0$  and  $a_1$  are uncertain about their observations. The situation is illustrated in Figure 7.

The next dawn, nobody left. On this day, the riddle is stuck. Given the visibility configuration, no agent can narrow down the set of possible worlds after the information they have received (and will receive). In the converging example, the agent with partial vision was able to reason through the perfect vision of the other agents. Now, there are two agents with the same partial vision. In other words, it is no longer possible to reason through the information and knowledge of two agents simultaneously, because at least one will have incomplete information.

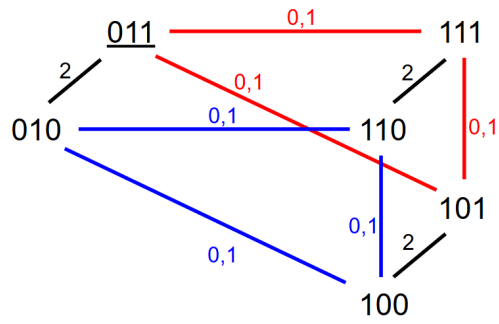


Figure 7: Formal Kripke model for the Blue-Eyed Islander riddle with a partial visibility configuration (agent 0 and agent 1 cannot see each other) after the first dawn. The actual world is underlined. Reflexive and symmetric relations are not drawn but are present. Colours in the model only serve visual purposes, as the partial vision models can become difficult to interpret.